

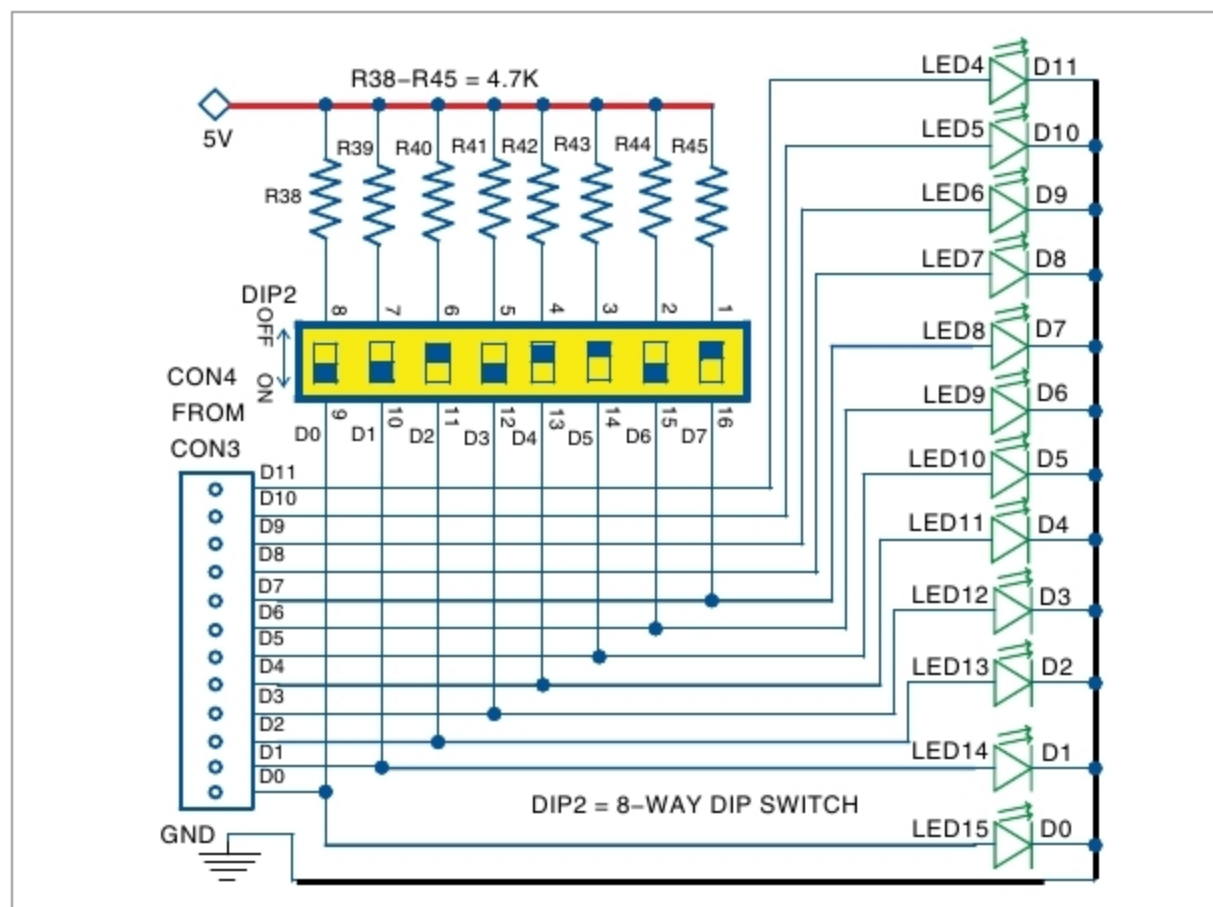


Fig. 4: Circuit diagram of I/O board

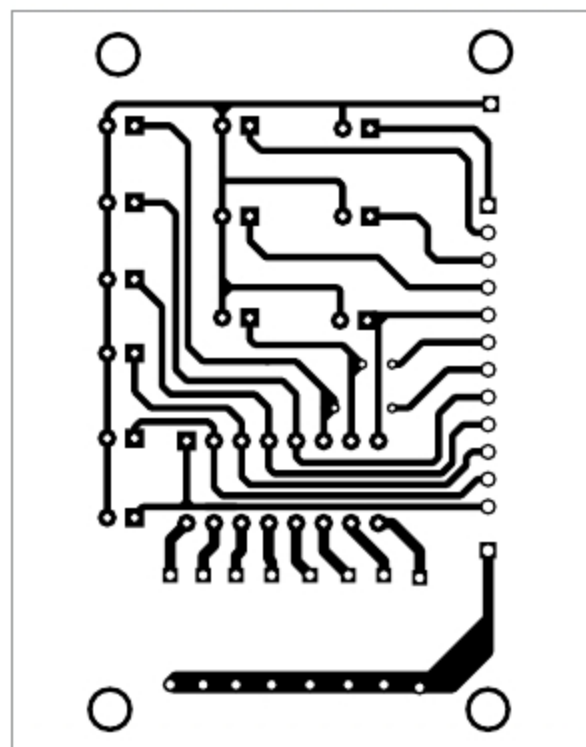


Fig. 5: Actual-size PCB layout of I/O board

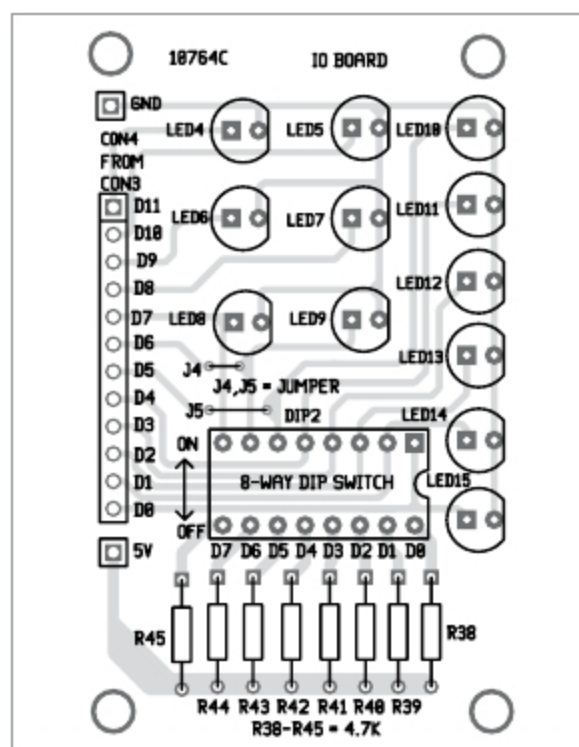


Fig. 6: Components layout for the PCB of I/O board

transmitter and receiver boards. Set transmitter board in TX mode with continuous data transmission by making SJ1 and SJ2 high. Set receiver board in RX mode without decoding by making SJ1 and SJ2 low.

Connect pin 3 of ATmega8A of transmitter board to pin 2 of ATmega8A of receiver board. Connect ground lines of both boards. Also, a data input-output (I/O) board can be connected temporarily to CON3 of transmitter and receiver boards for testing.

Circuit diagram of I/O board is shown in Fig. 4. PCB layout and com-

ponents layout of the I/O board are shown in Figs 5 and 6, respectively.

Now, switch on both the boards. LED2 indicates the status of the board setting, as shown in the table. Change address code A0-A3 and feed data input from transmitter side by ran-

PARTS LIST

(For each board with TX1 or RX1 along with I/O board)

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - Atmega8A MCU
- D1 - 1N4007 rectifier diode
- LED1-LED15 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R4 - 10-kilo-ohm
- R2, R5, R8-R11, R38-R45 - 4.7-kilo-ohm
- R3, R6, R7 - 2.2-kilo-ohm
- R12-R17, R32-R37 - 22-kilo-ohm
- R18 - 47-kilo-ohm
- R19-R31 - 3.3-kilo-ohm

Capacitors:

- C1 - 10 μ F, 16V electrolytic
- C2 - 100 μ F, 25V electrolytic
- C3, C4, C7 - 100nF ceramic disk
- C5, C6 - 22pF ceramic disk

Miscellaneous:

- XTAL1 - 3.6864MHz (optional)
- CON1 - 2-pin terminal connector
- CON2 - 6-pin connector
- CON3 - 12-pin male connector
- CON4 - 12-pin female connector
- TX1 - 433MHz RF transmitter module
- RX1 - 433MHz RF receiver module
- DIP1 - 4-way DIP switch
- DIP2 - 8-way DIP switch
- S1 - On/off switch
- S2 - Tactile switch
- SJ1, SJ2 - Shorting jumper
- Jumper wires
- 6-pin berg strip
- 9V DC-12V DC regulated power supply
- Heat-sink for 7805

domly connecting some data pins (D0-D11) to 5V through DIP switch. Then, observe the output on the 12 LEDs on receiver board. The same data should reflect on the LEDs on the receiver side, if address codes A0-A3 of the receiver are the same as transmitter address codes.

Once testing is successfully done, remove I/O board connections from both the boards. Connect 433MHz RX1 with RX board and 433MHz TX1 with TX board. The boards can now be tested for decoding and manual triggering as per the table. After successful testing, the boards are ready to use. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



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